sysMonitor AMQP/TLS/JSON

Interface Design Description

Service ID: *"sysMonitor"*

Abstract
This document describes the AMQP 0-9-1 protocol with TLS payload security and JSON payload encoding variant of the `sysMonitor` service of the GraphDBManager system (SysMonitor-AMQP-IDD-5.2.2).

Contents

1 Documentation edition and implementation scope **3**

2 Overview **3**
2.1 Interface registration . . . 3

3 Service Operations **4**
3.1 operation echo . . . 4
3.2 operation power . . . 4
3.3 operation status . . . 4

4 Data Models **6**
4.1 struct Observation . . . 6
4.2 Primitives . . . 6

5 Go and Rust implementation variants **7**
5.1 Go implementation . . . 7
5.2 Rust implementation . . . 7

6 References **7**

7 Revision History **8**
7.1 Amendments . . . 8
7.2 Quality Assurance . . . 8

1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the AMQP binding of the native `sysMonitor` service of GraphDBManager.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	AMQP	0-9-1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `sysMonitor` service interface

This document provides the Interface Design Description IDD to the *sysMonitor – Service Description* document [5].

2.1 Interface registration

With `sysmonitor.enabled=true`, GraphDBManager registers the service at start-up with interface template `generic_amqps` (protocol `amqps`, policy `CERT_AUTH`; `generic_amqp` without TLS) and the following properties and metadata:

```
{
  "serviceDefinitionName": "sysMonitor",
  "version": "1.0.0",
  "metadata": {
    "contract": "SysMonitor-AMQP-IDD-5.2.2",
    "schemaVersion": "1.0",
    "publisherMode": "native",
    "observationTypes": [
      "echo",
      "power",
      "status"
    ],
    "delivery": "persistent, publisher-confirmed"
  },
  "interfaces": [
    {
      "templateName": "generic_amqps",
      "protocol": "amqps",
      "policy": "CERT_AUTH",
      "properties": {
        "accessAddresses": [
```

```
          "amqp-broker"
        ],
        "accessPort": 5671,
        "exchange": "arrowhead",
        "baseRoutingKey": "TestCloud.sysmonitor.GraphDBManager.GraphDBManager",
        "operations": [
          "echo",
          "power",
          "status"
        ]
      }
    }
  ]
}
```

Listing 1: Registered service instance (TestCloud).

3 Service Operations

The publisher is the `SysMonitorPublisher` of the common library, configured with `sysmonitor.broker.address`, `sysmonitor.broker.port`, `sysmonitor.local.cloud` (default: from the system certificate), `sysmonitor.publish.interval` and `sysmonitor.ssl.*` (default: the `server.ssl` stores). The publisher connects to the broker (default `amqp-broker`, port 5671) with TLS, verifies the broker host name and authenticates with its X.509 system certificate (SASL `EXTERNAL`; the certificate CN is the broker user name). Every publication interval it publishes each observation to the topic exchange `arrowhead` with the routing key `<baseRoutingKey>.<operation>`, where the base routing key is `<Cloud>.sysmonitor.<System>.<serviceInstance>`. Messages are persistent (delivery mode 2), `application/json`, carry the observation identifier as message id, and are publisher-confirmed (timeout 5 s); a failed cycle is retried with a new connection at the next interval. The broker's topic permissions allow the system to publish only below its own base routing key.

3.1 PUBLISH `<Cloud>.sysmonitor.DataManager.DataManager.echo`

Operation: echo
Output: Observation

Liveness: `status` (`alive` or `degraded`), `uptimeSeconds` and a `detail` text; quality `good` or `uncertain`.

3.2 PUBLISH `<Cloud>.sysmonitor.DataManager.DataManager.power`

Operation: power
Output: Observation

Power supply: `supplyType=line`, `state=available`; unit `1`.

3.3 PUBLISH `<Cloud>.sysmonitor.DataManager.DataManager.status`

Operation: status
Output: Observation

Process state: `state=running`, `health` (`healthy`/`degraded`) and `serviceCount`, the number of registered services.

```
{
  "schemaVersion": "1.0",
  "observationId": "GraphDBManager-echo-1792670400000-42",
  "providerSystem": "GraphDBManager",
  "serviceInstance": "GraphDBManager",
  "observationType": "echo",
```

```
  "sourceTime": "2026-10-22T12:00:00.000Z",
  "sequence": 42,
  "value": {
    "status": "alive",
    "uptimeSeconds": 3600,
    "detail": "GraphDBManager is running"
  },
  "quality": "good",
  "metadata": {
    "publisherMode": "native"
  }
}
```

Listing 2: An echo observation.

4 Data Models

4.1 struct Observation

Object Field	Value Type	Description
"schemaVersion"	String	`1.0`.
"observationId"	String	`<System>-<type>-<epoch ms>-<sequence>`.
"providerSystem"	String	The publishing system.
"serviceInstance"	String	Service instance, default the system name.
"observationType"	String	`echo`, `power`, `status`.
"sourceTime"	String	ISO 8601 UTC time with milliseconds.
"sequence"	Number	Sequence number per publisher.
"value"	Object	Observed value, see the operations.
"unit"	String	`1` for `power` and `status`; absent for `echo`.
"quality"	String	`good` or `uncertain`.
"metadata"	Object	`{"publisherMode": "native"}`.

4.2 Primitives

Type	Description
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

5 Go and Rust implementation variants

5.1 Go implementation

No Go implementation folder for GraphDBManager is present in the designated primary Go tree. There is no port-specific executable, service binding or parity claim in this edition. A future port needs its own build, state model, security and interface qualification.

5.2 Rust implementation

No Rust implementation folder for GraphDBManager is present in the designated primary Rust tree. There is no port-specific executable, service binding or parity claim in this edition. A future port needs its own build, state model, security and interface qualification.

6 References

[1] AMQP Working Group, “Amqp 0-9-1 specification,” AMQP Working Group, Tech. Rep., 2008.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “sysMonitor – Service Description (GraphDBManager),” Eclipse Arrowhead, SD 5.3.0, 2026, graphDBManager-v5.3.0-sysMonitor-SD.pdf.

7 Revision History

7.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-28	5.2.2	Document created from the corresponding DataManager document for the GraphDB-backed copy GraphDBManager.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

7.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	